DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

Design and Performance Comparison of Low-Power 6T, 8T and 10T SRAM Cells

MAIN PROJECT | ANALOG / MEMORY DESIGN

Prepared for: VLSI Design Project Submission

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Academic status: ECE, 3rd semester

Report status: Proposed design and verification plan. Numerical simulation results are not claimed until tools are run.

Abstract

This main project proposes design and performance comparison of 6T, 8T and 10T static random-access memory (SRAM) cells for low-power VLSI. A common testbench will evaluate hold stability, read disturbance, write ability, power and access delay under a specified transistor model.

1. Problem statement

Conventional 6T SRAM is compact but can exhibit a read-stability versus write-ability trade-off at low supply voltages. Additional-transistor cells can decouple read paths or improve assist options, at a potential area cost. Compare selected 6T, 8T and 10T cell schematics under consistent operating conditions.

2. Objectives

- Construct working transistor-level schematics for one explicitly defined 6T, 8T and 10T topology.
- Demonstrate hold, read-0, read-1, write-0 and write-1 operations.
- Estimate static noise margin (SNM), read/write delay, leakage and dynamic energy.
- Discuss area, bitline loading and peripheral-circuit trade-offs.

3. Literature review

A conventional 6T cell consists of two cross-coupled inverters and two access devices. An 8T cell commonly adds a separate read port, reducing read disturbance. There are multiple incompatible 10T architectures; the chosen 10T topology must be identified by its actual schematic and source, not only its transistor count. Relevant literature addresses read stability, write ability, low-voltage operation and leakage reduction.

4. SRAM architecture diagram

The simplified 6T diagram below shows the bistable storage pair, two access transistors and complementary bitlines. It is a connectivity overview rather than a foundry-ready transistor schematic.

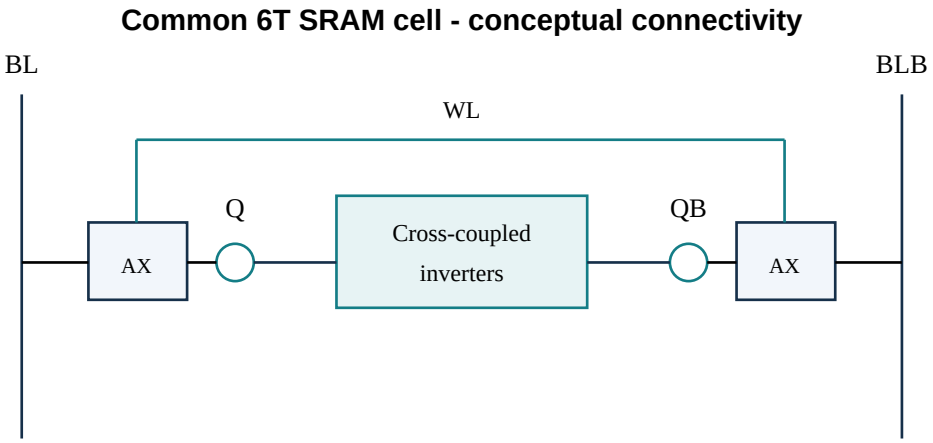

Figure 2. Simplified 6T cell: storage pair + two access transistors

5. Cell operation

Hold: WL low, storage nodes retain complementary states. Read: precharge bitlines, assert wordline, detect a differential bitline change. Write: drive complementary bitlines and assert WL long enough to overturn the latch. In an 8T or 10T topology, read/write ports may differ and must be modeled separately.

6. Proposed experimental setup

Use a common PDK/model, device corner, temperature, VDD sweep and extracted/equivalent load assumptions. For each cell, include a realistic read path and precharge/write stimulus. Measure read access delay to a defined bitline voltage difference, write delay to node switching, hold leakage and read/write energy.

7. Evaluation metrics

Metric	Method / definition
Hold SNM	Butterfly curves from cross-coupled inverter DC transfer characteristics
Read SNM	DC noise-margin measurement under read bias
Write ability	Write trip point or minimum write voltage / write margin
Read delay	WL activation to specified sensed bitline differential
Write delay	WL activation to internal node crossing criterion
Energy	Integral of VDD x supply current over operation
Area proxy	Cell footprint from layout; transistor count alone is insufficient

8. Comparison matrix - experimental results pending

Metric	6T	8T	10T
Transistor count	6	8	10
Hold SNM	TBD	TBD	TBD
Read SNM	TBD	TBD	TBD
Write delay	TBD	TBD	TBD
Read energy	TBD	TBD	TBD
Cell area	TBD	TBD	TBD

9. Workflow and milestones

• Week 1: Review SRAM cell fundamentals and choose traceable 8T/10T schematics.

• Week 2: Draw and functionally test 6T cell.

• Week 3: Build and test 8T and 10T topologies.

• Week 4: Run hold/read/write, SNM and voltage-sweep experiments.

• Week 5: Tabulate results and produce waveforms and plots.

• Week 6: Write limitations, final discussion and demonstration.

10. Design risks and mitigations

Read stability can depend on access/pull-down ratios, transistor sizing and bitline precharge. Write ability depends on pull-up/access ratios and write drivers. Peripheral circuits, corners and mismatch can change the ranking. For meaningful claims, use the same process and clearly report the selected 10T architecture.

11. Expected outcome

A repeatable comparative framework for SRAM cell stability, performance and energy. The relative winner is a measured conclusion, not assumed in advance.

12. Conclusion

The main project offers a practical bridge from transistor-level CMOS design to semiconductor memory engineering, including reliability and low-voltage design trade-offs.

13. References and further reading

J. M. Rabaey et al., Digital Integrated Circuits; N. H. E. Weste and D. Harris, CMOS VLSI Design. Include verified primary literature identifying the exact 8T and 10T variants selected.